NOSQL DATABASE DESIGN – LAB SHEET - 4

1. Install and configure Apache HBase. Create a table named `Student` with column families `personal` and `academic`. Insert at least five student records and display them using HBase shell commands.

2. Using Apache HBase, perform basic CRUD operations on an `Employee` table. Demonstrate `create`, `put`, `get`, `scan`, and `delete` operations.

3. Design an HBase table for an Event Logging System. Store `event_id`, timestamp, user information, event type, and status using appropriate column families. Insert sample events and retrieve them based on row keys.

4. Create an HBase table for a Content Management System (CMS) with column families such as `content`, `author`, and `metadata`. Insert, update, retrieve, and delete content records.

5. Implement a simple Blogging Platform using HBase. Create suitable column families for blog details, author information, and comments. Insert at least three blog records and retrieve a particular blog using its row key.

6. Demonstrate the column-family data model in HBase by creating a table with multiple column families and different qualifiers. Explain through your implementation how HBase stores columns within column families.

7. Create an HBase table named `WebsiteCounter` to maintain page-view counters. Implement increment operations and display the updated counter values for different pages.

8. Implement expiring usage/session data using HBase. Design a table for user sessions and configure a TTL (Time-To-Live) for the required column family. Insert session records and verify their expiration behavior.

9. Design an HBase table for a large-scale application and explain how its row-key design can support horizontal scaling. Demonstrate insertion of records using a suitable row-key strategy.

10. Create an HBase table with multiple regions and demonstrate how HBase distributes data across regions. Explain the role of RegionServers in achieving scalability and availability.

11. Install and configure Apache Cassandra. Create a keyspace named `CollegeDB` with an appropriate replication factor and create a `Students` table. Insert and retrieve at least five student records.

12 .Using Apache Cassandra, perform CRUD operations on an `Employees` table. Demonstrate `INSERT`, `SELECT`, `UPDATE`, and `DELETE` operations using CQL.

13. Design a Cassandra database for an event logging application. Create an appropriate table, insert multiple events, and write CQL queries to retrieve events for a particular user and time period.

14. Demonstrate availability and consistency in Cassandra by creating a keyspace with a suitable replication strategy. Perform read/write operations using different consistency levels such as `ONE`, `QUORUM`, and `ALL`, and observe the behavior.

15 .Design and implement a column-oriented NoSQL application using HBase or Cassandra for any one of the following: Event Logging, Content Management, Blogging Platform, Counters, or Expiring Usage. Demonstrate database/table creation, data insertion, querying, updating, deletion, and explain how consistency, availability, transactions, query features, and scaling are handled by the selected database.